Quantitative Data Analysis

Session 3 — Comparing Groups: ANOVA & Post-hoc Tests

NGUYEN Anh-Tuan
EM Normandie — Programme Grande École
Course code 2627_ECO_2_EN_009 · 2026–2027

Where we are

Session	Method	Question it answers
1	Univariate descriptive analysis	What does my data look like?
2	Chi-square, correlation & t-tests	Are two variables related? Do two groups differ?
3	**ANOVA + post-hoc tests**	**Do groups differ from each other?**
today		
4	Simple & multiple regression	What drives my outcome?

Last week: two qualitative variables (χ^2), two quantitative ones (r), and **two groups** (Student or Welch t-test).
Today, the general mixed case: a quantitative outcome compared across **three or more groups** of a qualitative variable.

Today: 3 hours 30

Time	What happens	Mode
0:00 – 0:15	Recap: the three most common mistakes of Session 2	Plenary
0:15 – 1:00	**Theory:** why ANOVA, the logic of variance, the F test	Lecture + quiz
1:00 – 1:25	**Conditions** and what to do when they fail	Lecture
1:25 – 1:40	*Break*	
1:40 – 2:25	**Post-hoc tests:** which groups differ — and which test to use, on four datasets NEW	Lecture
2:25 – 3:00	**Jamovi lab:** one ANOVA on your own dataset	Teams of 3
3:00 – 3:30	Write-up of your results in protocol format	Checkpoint

> You leave today knowing not only **whether** your groups differ, but **which ones** — which post-hoc test to trust — and what a manager should do about it.

Why ANOVA?

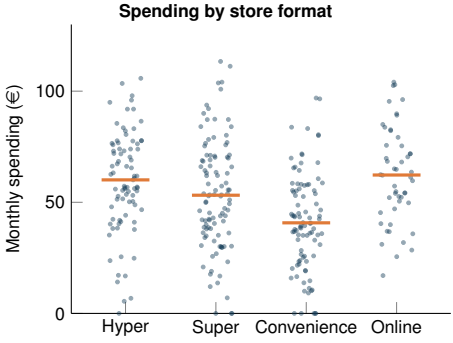
The question

Same retail data ($N = 325$). Customers shop mainly in one of four store formats. Does monthly spending depend on the format?

Outcome (Y): monthly spending €
quantitative, continuous

Factor (X): store format
nominal, 4 groups

Quantitative × qualitative (3+ groups)
→ **one-way ANOVA + post-hoc**



Two groups or more than two?

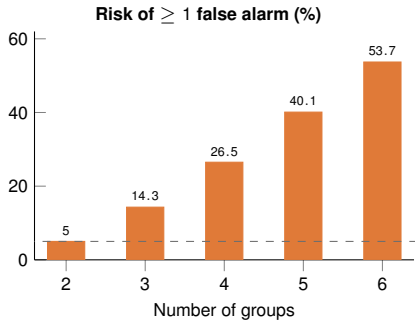
2 groups (e.g. card holders vs non-holders)
$\rightarrow$ independent-samples t-test (Session 2)

3 groups or more (e.g. 4 store formats)
$\rightarrow$ one-way ANOVA, **one single test** for all groups at once

Why not six t-tests? Each test carries a 5% risk of a false alarm. The risks add up:

$$P(\geq 1 \text{ false alarm}) = 1 - 0.95^m$$

m = number of pairs $= k(k-1)/2$.



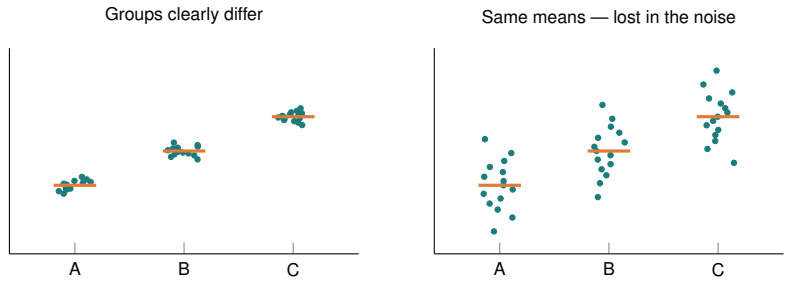
ANOVA keeps the overall risk at 5%, whatever the number of groups.

From the t-test to ANOVA: the same family NEW

Situation	2 groups (Session 2)	3+ groups (today)
Equal variances (Levene $p > .05$)	Student's t-test	**Fisher's ANOVA** + **Tukey**
Unequal variances (Levene $p \leq .05$)	Welch's t-test	**Welch's ANOVA** + **Games–Howell**
Small groups, not normal	Mann–Whitney U	**Kruskal–Wallis** + **DSCF**
Effect size	Cohen's *d*	η^2

Everything you learned about Student vs Welch carries over: **Fisher's ANOVA and Tukey pool one common variance**; Welch's ANOVA and Games–Howell keep each group's own. When spreads and group sizes differ, pooling gives false alarms or missed effects — exactly as with the t-test.

The idea: compare two kinds of variation



Between-group variation: how far the group means (orange) are from each other. *Identical in both panels.*

Within-group variation: how spread out customers are inside each group. *Small on the left, large on the right.*

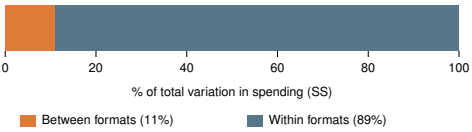
The F ratio

$$F = \frac{\text{variation between groups}}{\text{variation within groups}} = \frac{MS_{between}}{MS_{within}}$$

- $F \approx 1$: the means differ no more than individuals do $\rightarrow$ **noise**
- $F \gg 1$: the gap between groups stands out $\rightarrow$ **signal**

The name "analysis of variance" is misleading: it is a test on **means**, done by comparing variances.

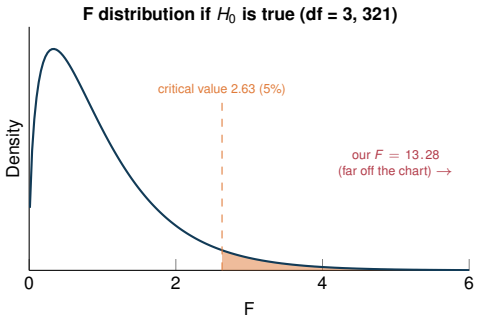
Our case: total variation split in two



$$SS_{total} = SS_{between} + SS_{within}$$

$$200\,661 = 22\,151 + 178\,510$$

From F to a p-value



If H_0 were true, F would usually be close to 1.
Beyond 2.63, it happens less than 5% of the time.

Our $F = 13.28 \rightarrow p < .001$.

> The software does this for you. What you must be able to say: "**the gap between formats is far larger than the gap between customers of the same format.**"

Hypotheses — and a classic trap

H_0: average spending is the **same** in all four store formats.

$\mu_{Hyper} = \mu_{Super} = \mu_{Conv} = \mu_{Online}$

H_1: **at least one** format has a different average spending.

Trap: H_1 is **not** "all the formats are different". A significant ANOVA only says the four means are not all equal. It does not say which ones differ.



Quiz — ANOVA, t-test or something else?

1. ROI (%) of firms in 5 sectors.
2. Satisfaction of customers who did / did not use click-and-collect.
3. Share of women (%) among the managers of 3 subsidiaries.
4. Emissions per employee across ESG grades A, B, C, D.
5. Brand purchased (national / regional) across 4 regions.
6. Basket size before and after a store refit, same 60 stores.
7. Sales of 12 stores in 3 regions (4 per region), very skewed.
8. Turnover and number of employees of 300 firms.

Take 3 minutes. What is the outcome? What is the grouping variable?

Quiz — answers

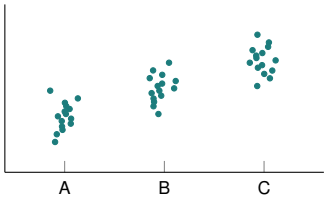
#	Test	Why / trap
1	One-way ANOVA + post-hoc	Quantitative outcome, 5 groups.
2	Independent-samples t-test	Only 2 groups (ANOVA gives the same p, but the t-test is the convention). Levene → Student or Welch.
3	Chi-square	The "%" hides the real data: each manager is a woman or a man → two qualitative variables.
4	One-way ANOVA + post-hoc	ESG grade is ordinal, but used here as 4 groups.
5	Chi-square	The outcome is qualitative. ANOVA needs a quantitative Y.
6	Paired-samples t-test	The same stores twice: not independent groups.
7	Kruskal–Wallis (+ DSCF)	Tiny groups and skewed data: non-parametric version of ANOVA.
8	Correlation (Session 2)	Two quantitative variables, no groups.

Conditions: can I trust the F test?

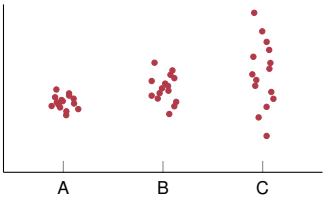
Three conditions

1. **Independence** — each customer belongs to one group only, and is measured once.
2. **Normality within each group** — or at least 30 observations per group.
3. **Homogeneity of variances** — the spread is similar in all groups. Tested with **Levene's test**.

Homogeneous: similar spreads



Heterogeneous: one group spreads out



Levene's test — the opposite logic again

H_0: the variances are **equal** in all groups.
So, like Shapiro–Wilk in Session 1, $p > 0.05$ **is the good news**.

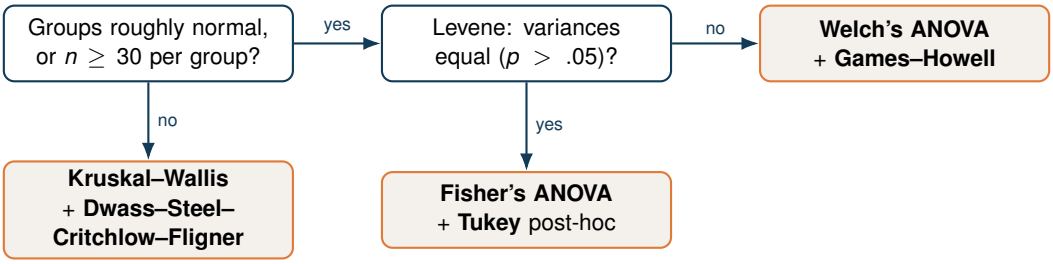
Homogeneity of Variances Test (Levene's)

F	df1	df2	p
0.52	3	321	0.666

$p = 0.666 > 0.05 \rightarrow$ we keep H_0: variances are homogeneous. **Classic (Fisher) ANOVA is fine.**

Normality: Shapiro–Wilk is significant in three groups (skewness $\approx$ 0.3), but every group has $n \geq 49$. With groups this large, ANOVA is robust — and we will confirm with a non-parametric test.

When the conditions fail: a decision path



Jamovi runs Welch by default in **ANOVA** → **One-Way ANOVA**. Good practice: tick both Welch and Fisher. If they agree, the conclusion is robust.

Our case: Fisher $F(3, 321) = 13.28$; Welch $F(3, 158) = 14.34$; Kruskal–Wallis $\chi^2(3) = 33.99$ — all $p < .001$.

Why Welch's ANOVA exists NEW

Fisher's F divides by MS_{within} — **one pooled variance** for all groups, weighted by group size. Same weakness as Student's t:

- small groups with large spread → F **overstated**
- large groups with large spread → F **understated**

Welch's ANOVA weights each group by n_j / s_j^2 and corrects the second df (a decimal number).

Example (Dataset 2 below): 4 channels, SDs from 8 to 30 €, n from 20 to 120. Levene $F(3, 221) = 37.81$, $p < .001$.

Test	F	p
Fisher	$F(3, 221) = 5.99$	$< .001$
Welch	$\mathbf{F(3, 48.7) = 4.21}$	**.010**

Both are significant here, but Fisher overstates the evidence by a factor of 10 in p. With a smaller effect, Fisher alone could give a false "significant". A decimal second df in the output = Welch.

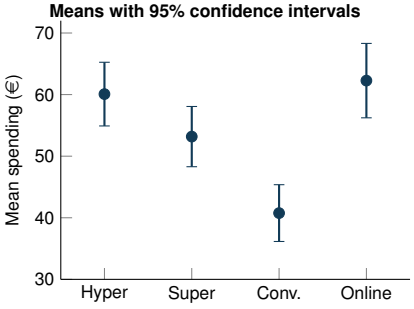
Running and reading the ANOVA

Step 1: describe the groups

Group Descriptives

Format	N	Mean	SD	SE
Hypermarket	83	60.08	24.01	2.64
Supermarket	101	53.18	25.05	2.49
Convenience	92	40.76	22.52	2.35
Online	49	62.27	21.59	3.08

Standard deviations are close (21.6 to 25.1): consistent with Levene. Convenience customers spend about **20 € less** per month. SE $= SD/\sqrt{n}$.



Step 2: the ANOVA table

ANOVA — Monthly spending (Jamovi, ANOVA module)

	Sum of Squares	df	Mean Square	F	p	η^2
Store format	22 151	3	7 384	13.28	$< .001$	0.110
Residuals	178 510	321	556			

How to read it:

- $df_{between} = k - 1 = 4 - 1 = 3$
- $df_{within} = N - k = 325 - 4 = 321$
- $MS = SS/df$ and $F = 7\,384/556 = 13.28$
- Report as: $F(3, 321) = 13.28$, $p < .001$

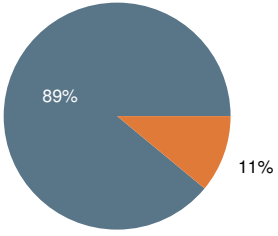
$p < 0.05 \rightarrow$ we reject H_0.
Average spending is **not the same** in all store formats.

Step 3: how much does it matter? η^2

$$\eta^2 = \frac{SS_{between}}{SS_{total}} = \frac{22\,151}{200\,661} = 0.11$$

Store format accounts for **11%** of the differences in spending between customers.

η^2	Effect
≈ 0.01	small
≈ 0.06	medium
≥ 0.14	large

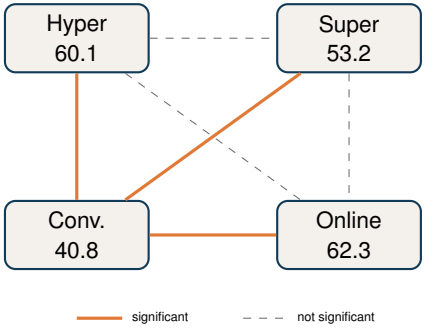


Store format (orange) vs everything else

A **medium** effect: format matters, but 89% of the variation lies elsewhere — visits, income… Session 4 will put them in one model.

Post-hoc tests: which groups differ?

Four groups, six comparisons



With k groups there are $k(k-1)/2$ pairs: **6 pairs** here.

Post-hoc tests compare every pair while **correcting the p-values**, so that the overall risk stays at 5%.

> **The picture:** one format stands apart. Every link to Convenience is significant; the other three formats do not differ from each other.

Which post-hoc test?

Test	When	How it corrects	Comment
Tukey	Equal variances (Levene $p > .05$)	Pooled MS_{within} for every pair; studentized range distribution, built for "all pairs at once"	The standard choice after a Fisher ANOVA
Games–Howell	Unequal variances	Each pair gets its own SE $\sqrt{s_1^2/n_1 + s_2^2/n_2}$ and Welch df	The partner of Welch's ANOVA
Bonferroni	Few **planned** comparisons	$p_{adj} = \min(1,\ p \times m)$, m = number of comparisons	Very conservative: multiplies each p by the number of pairs
Dwass–Steel–Critchlow–Fligner	After Kruskal–Wallis	Ranks each pair of groups together (Wilcoxon), then studentized range	Non-parametric pairwise comparisons

Only run post-hoc tests after a **significant** ANOVA. If $p \geq .05$ overall, there is nothing to locate — stop there and report it.

Next: four datasets, one per test, showing what goes wrong with the wrong choice.

Why correct at all? A simulation NEW

4 000 datasets in which all group means are **truly equal** ($n = 30$ per group). How often does at least one pair come out “significant”? Every one of those is a false alarm.

Groups	Pairs	Uncorrected	Tukey	Bonferroni	$1 - 0.95^m$
3	3	**11.9%**	4.8%	3.9%	14.3%
4	6	**19.1%**	4.5%	3.3%	26.5%
5	10	**29.1%**	5.5%	4.3%	40.1%

Uncorrected t-tests raise false alarms **2 to 6 times** too often.

Tukey sits on 5%.
Bonferroni stays under 5%: safe but over-cautious, which costs power.

The real rate is below the formula $1 - 0.95^m$ because pairs share groups — but it is still far above 5%.

Dataset 1 — equal variances $\rightarrow$ Fisher + Tukey

Data: monthly spending by store format (the case above). Levene $p = .666 \rightarrow$ Fisher's ANOVA $F(3, 321) = 13.28$, $p < .001 \rightarrow$ **Tukey**.

Post Hoc Comparisons — Store format (Tukey)

Comparison		Mean diff.	SE	t	p_{tukey}
Hyper	Super	6.90	3.49	1.97	0.200
Hyper	Conv.	19.32	3.57	5.41	$< .001$
Hyper	Online	−2.19	4.25	−0.52	0.955
Super	Conv.	12.43	3.40	3.66	0.002
Super	Online	−9.09	4.11	−2.21	0.122
Conv.	Online	−21.51	4.17	−5.16	$< .001$

Mean diff. = first group minus second group. The sign tells you who is higher.

Significant: Convenience spends less than Hypermarket (−19.3 €), Supermarket (−12.4 €) and Online (−21.5 €).

Not significant: Hyper, Super and Online. The 9 € gap Super–Online is not a proven difference ($p = .122$).

Dataset 1 — what an uncorrected analysis would have claimed NEW

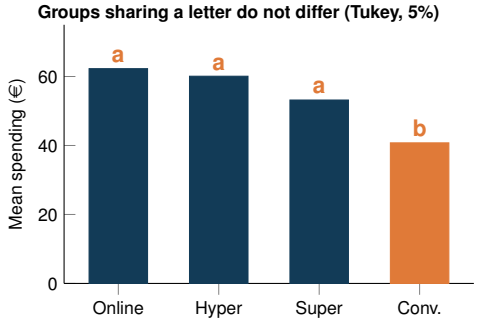
Pair		Mean diff.	p uncorrected	p **Tukey**	p Bonferroni ($\times 6$)
Hyper	Super	6.90	**0.049**	**0.200**	0.295
Hyper	Conv.	19.32	$< .001$	$<$ **.001**	$< .001$
Hyper	Online	-2.19	0.607	**0.955**	1.000
Super	Conv.	12.43	$< .001$	**0.002**	0.002
Super	Online	-9.09	**0.028**	**0.122**	0.165
Conv.	Online	-21.51	$< .001$	$<$ **.001**	$< .001$

Computed on a dataset rebuilt with the slide's group means and SDs.

Why correct: six uncorrected t-tests would "find" Hyper $>$ Super and Online $>$ Super. With six tries, a p just under .05 is what chance produces. Tukey sends both back.

Why Tukey, not Bonferroni: same verdicts, but Bonferroni's p are larger (0.295 vs 0.200; 0.165 vs 0.122). Tukey is built for all pairs; Bonferroni wastes power there.

Summarise the result in one picture



The letter display is the fastest way to show a post-hoc result to a manager:

- same letter = no significant difference
- different letters = significant difference

$$\text{Online} \approx \text{Hyper} \approx \text{Super} > \text{Convenience}$$

Dataset 1 — writing it up (model answer)

1. IDENTIFY. Monthly spending is quantitative and store format is a nominal variable with four groups: one-way ANOVA with post-hoc tests. Groups are independent; Levene's test is not significant ($F(3, 321) = 0.52$, $p = .666$); every group has $n \geq 49$, so ANOVA is robust to the mild skewness.

2. HYPOTHESISE. H_0: average spending is the same in the four formats. H_1: at least one format differs.

3. DECIDE. $F(3, 321) = 13.28$, $p < .001$: we reject H_0. $\eta^2 = 0.11$, a medium effect. Tukey tests show that Convenience customers spend significantly less (40.8 €) than Hypermarket (60.1 €), Supermarket (53.2 €) and Online (62.3 €) customers; the other three formats do not differ. Welch and Kruskal–Wallis confirm.

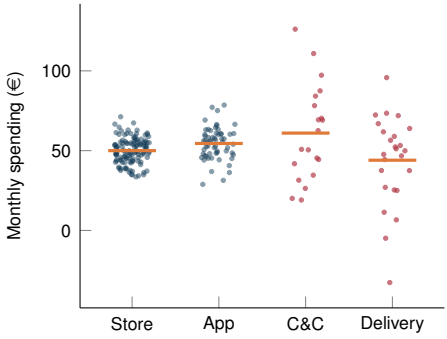
4. RECOMMEND. Convenience is the under-performing format: raise its basket with larger-pack promotions and cross-selling at the till, rather than spreading a single promotional budget across all four formats. Do not invest to "close the gap" between Supermarket and Online: that gap is not established.

Dataset 2 — unequal variances: the data NEW

Monthly spending by purchase channel. Store and app customers are numerous and homogeneous; click-and-collect and delivery customers are few and very heterogeneous.

Channel	n	Mean (€)	SD (€)
Store	120	50.00	8.00
App	60	54.50	10.00
Click & collect	20	61.00	**30.00**
Home delivery	25	44.00	**28.00**

Levene $F(3, 221) = 37.81$, $p < .001 \rightarrow$ variances **not** equal $\rightarrow$ **Welch's ANOVA**: $F(3, 48.7) = 4.21$, $p = .010 \rightarrow$ **Games–Howell**.



Dataset 2 — Tukey and Games–Howell disagree NEW

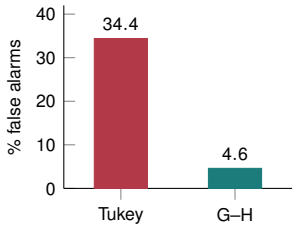
Pair		Mean diff.	Tukey SE	p Tukey	G–H SE	p **Games–Howell**
Store	App	−4.50	2.36	0.230	1.48	**0.016**
Store	Click & collect	−11.00	3.61	**0.014**	6.75	**0.386**
Store	Delivery	6.00	3.29	0.264	5.65	**0.715**
App	Click & collect	−6.50	3.86	0.335	6.83	**0.778**
App	Delivery	10.50	3.56	**0.018**	5.75	**0.283**
Click & collect	Delivery	17.00	4.49	**0.001**	8.74	**0.226**

Tukey: 3 false alarms. It uses one pooled SD ($\sqrt{MS_{within}} = 15.0\,€$) for every pair — **too small** for the pairs involving C&C and delivery (own SDs 28–30 €, only 20–25 customers).

Tukey: 1 real effect missed. The same pooled SD is **too large** for Store vs App (SDs 8 and 10 €, 180 customers). Their 4.5 € gap is measured precisely: only Games–Howell sees it.

Dataset 2 — proof by simulation, and the write-up

Same n and SDs, **equal** true means, 3 000 datasets: how often is at least one pair "significant"?



1. IDENTIFY. Spending is quantitative, channel nominal with 4 independent groups: one-way ANOVA. Levene is significant ($F(3, 221) = 37.81$, $p < .001$) and group sizes are very unequal → Welch's ANOVA + Games–Howell.
2. HYPOTHESISE. H_0: average spending is the same in all channels. H_1: at least one differs.
3. DECIDE. Welch $F(3, 48.7) = 4.21$, $p = .010$: reject H_0. Games–Howell: app customers spend 4.5 € more than store customers ($p = .016$). Larger gaps involving C&C and delivery (up to 17 €) are not significant: those customers are few and very dispersed.
4. RECOMMEND. Push store customers to the app (app-only offers): the only established gain, +4.5 €/month. Do not reallocate budget to click-and-collect on its high average; collect more data first.

Dataset 3 — a few planned comparisons: the data NEW

A retailer tests **four new store layouts** (A–D) against the current one (control), 30 stores each, average basket (€).

> **Before collecting data**, management decided it only cares about two questions:
> (1) is layout A better than control?
> (2) is layout B better than control?

Levene $F(4, 145) = 0.08$, $p = .99$. Fisher $F(4, 145) = 3.90$, $p = .005$, $\eta^2 = 0.10$.

Layout	n	Mean (€)	SD (€)
Control	30	50.00	12.00
A	30	58.00	12.00
B	30	51.00	12.00
C	30	55.00	12.00
D	30	47.00	12.00

5 groups $\rightarrow$ 10 possible pairs. But only **2** were planned.

Dataset 3 — Bonferroni on the planned comparisons NEW

Comparison	Mean diff.	p uncorrected	**Bonferroni, 2 planned ($\times$2)**	Tukey, all 10 pairs	Bonferroni, all 10 ($\times$10)
A – Control	8.00	0.011	**0.022 $\rightarrow$ significant**	0.079	0.108
B – Control	1.00	0.747	**1.000**	0.998	1.000

Bonferroni's penalty is the **number of comparisons** m. With 2 planned comparisons each p is only doubled: layout A's +8 € is significant ($p = .022$).

Tukey protects all 10 pairs — which nobody asked about — and misses A ($p = .079$). Bonferroni over all 10 pairs is stricter still ($p = .108$): hence "very conservative" with many pairs.

The rule that makes this honest: the comparisons are chosen **before** seeing the data. Looking at the means, picking the biggest gap and "planning" it afterwards is cheating — the real number of comparisons was 10.

Dataset 3 — writing it up (model answer)

1. IDENTIFY. Basket value is quantitative, layout is nominal with five independent groups: one-way ANOVA. Levene is not significant ($p = .99$). Two comparisons against control were planned before the study, so we test them with a **Bonferroni correction ($m = 2$)**.

2. HYPOTHESISE. H_0: layout A (resp. B) has the same average basket as the current layout. H_1: it differs.

3. DECIDE. The ANOVA is significant ($F(4, 145) = 3.90$, $p = .005$, $\eta^2 = 0.10$). Layout A raises the average basket by 8 € (58 vs 50 €; Bonferroni $p = .022$). Layout B does not differ from control ($p = 1.00$).

4. RECOMMEND. Roll out layout A (+8 € per basket, about +16%); drop layout B. Layouts C and D were not part of the plan: any idea about them is a new hypothesis for a new study.

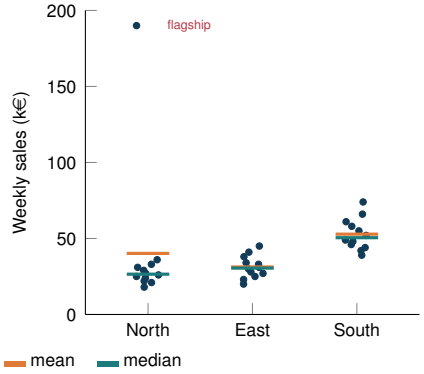
Among all 10 pairs, Tukey finds only A vs D significant (+11 €, $p = .005$) — an unplanned finding to be confirmed later.

Dataset 4 — skewed data and an outlier: the data NEW

Weekly sales (k€) of 12 stores per region. North includes one flagship store at 190 k€.

Region	Mean	**Median**	SD	Shapiro–Wilk
North	40.17	**26.5**	47.46	$W = 0.43$, $p < .001$
East	31.25	**30.5**	7.41	$p = .98$
South	52.83	**50.5**	10.41	$p = .74$

Groups of 12 (< 30) and North is far from normal (skewness 2.95) $\rightarrow$ the decision path leads to **Kruskal–Wallis + DSCF**.



Dataset 4 — ANOVA sees nothing, Kruskal–Wallis does NEW

Omnibus test	Result	Verdict
Fisher's ANOVA	$F(2, 33) = 1.75$, $p = .19$	**n.s.**
Kruskal–Wallis	$\chi^2(2) = 18.80$, $p < .001$	**Significant**

Pair	p Tukey	DSCF W	p **DSCF**
North – East	0.724	-1.55	**0.515**
North – South	0.525	-4.90	**0.002**
East – South	0.166	-5.55	$<$ **.001**

The 190 k€ store drags North's mean to 40.2 (median 26.5) and inflates the within-group variance: $MS_{within} = 805$ with it, 65 without it. The noise swamps the signal.

Ranks do not care how far 190 is from the rest — only that it is the largest. DSCF sees what the medians show: **South sells more than both other regions**; North and East do not differ.

Do not simply delete the flagship: it is a real store. Choosing the rank-based test is the honest fix. Effect size: $\varepsilon^2 = H/(N-1) = 18.80/35 = 0.54$.

Dataset 4 — writing it up (model answer)

1. IDENTIFY. Weekly sales are quantitative and region is nominal with three independent groups. Groups are small ($n = 12$) and North is strongly skewed by one flagship store (Shapiro–Wilk $p < .001$), so we use **Kruskal–Wallis with DSCF** pairwise comparisons.

2. HYPOTHESISE. H_0: sales have the same distribution in the three regions. H_1: at least one region tends to sell more or less.

3. DECIDE. $\chi^2(2) = 18.80$, $p < .001$, $\varepsilon^2 = 0.54$: we reject H_0. DSCF shows that South (median 50.5 k€) sells more than North (26.5 k€, $p = .002$) and East (30.5 k€, $p < .001$); North and East do not differ ($p = .515$).

4. RECOMMEND. Study what South stores do differently (range, staffing, location) and transfer it to North and East. Analyse the North flagship separately: it is a different kind of store and should not set targets for the other North stores.

With a rank-based test, report **medians**, not means.

What the four datasets teach NEW

Dataset	Condition that decides	Right tests	What the wrong test did
1. Store formats	Levene $p > .05$, $n \geq 49$	Fisher + **Tukey**	Uncorrected t-tests: 2 false differences. Bonferroni: same verdicts, less power
2. Channels	Levene $p < .001$, unequal n	Welch + **Games–Howell**	Tukey: 3 false differences, 1 real one missed
3. Layouts	2 comparisons planned in advance	**Bonferroni** ($m = 2$)	Tukey on all 10 pairs: misses layout A's +8 €
4. Regional sales	Small n, strong skew, outlier	Kruskal–Wallis + **DSCF**	Fisher + Tukey: nothing significant at all

The post-hoc test is **not a matter of taste**: it follows from the same checks that chose the ANOVA — normality or group size, then Levene — plus one question: were the comparisons planned?

Read output critically — a real example

ANOVA: customer satisfaction (0–20) by region — North, South, East, West ($N = 200$)

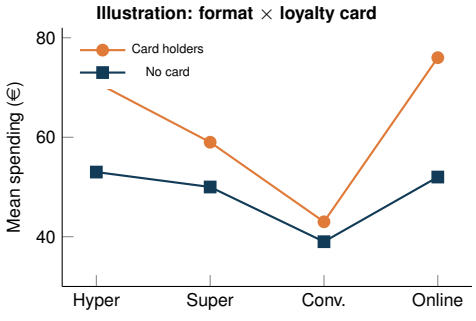
	Sum of Squares	df	Mean Square	F	p
Region	120	4	40.0	12.4	$< .001$
Residuals	1 764	196	9.0		

Two things are wrong here.
Can you spot them before I tell you?

df = 4 — four regions give $df = k - 1 = 3$. (And $3 + 196 = 199 = N - 1$: the table itself says so.)
F = 12.4 — $F = 40.0/9.0 = 4.44$. The F printed does not come from these mean squares.

Check df, check $F = MS_{between}/MS_{within}$. Thirty seconds, every time.

Going further: two factors at once



A **two-way ANOVA** tests two factors and their **interaction**.

Interaction = the effect of one factor depends on the other. Here the card matters a lot online and in hypermarkets, much less in convenience stores: **the lines are not parallel**.

> Beyond this course — Jamovi does it in **ANOVA → ANOVA** by adding a second fixed factor.

Jamovi lab

One-way ANOVA in Jamovi

Analyses → ANOVA → One-Way ANOVA

1. **Dependent Variables:** your quantitative outcome
2. **Grouping Variable:** your nominal variable (3–5 groups)
3. **Variances:** tick Don't assume equal (Welch's) **and** Assume equal (Fisher's)
4. **Additional Statistics:** Descriptives table and Descriptives plots
5. **Assumption Checks:** Equality of variances, Normality
6. **Post-Hoc Tests:** Tukey (equal variances) or Games–Howell (unequal)

Non-parametric: ANOVA → Non-Parametric → One-Way ANOVA (Kruskal–Wallis); tick **DSCF pairwise comparisons**.
Bonferroni: ANOVA → ANOVA → Post Hoc Tests, correction "Bonferroni".

Read, in this order:

1. Descriptives + plot — the story
2. Levene — which ANOVA?
3. F, df, p — is there a difference?
4. Post-hoc — where is it?
5. η^2 — does it matter?

η^2 is not in One-Way ANOVA: use **ANOVA → ANOVA** and tick η^2 under Effect Size.

Apply it to your fil rouge dataset

Firm-level data
sector → ROI
ESG grade band → emissions per employee
size band → net margin

Retail / customer data
region → monthly spending
store format → satisfaction score
socio-professional category → loyalty score

Before running anything:

- Is the grouping variable set to **Nominal** (or Ordinal)?
- Are there **3–5 groups**? Merge tiny ones (e.g. $n < 10$) into "Other".
- Plot the outcome by group (Descriptives → split by, box plot).
- Compare the **SDs and group sizes**: very different? Think Welch + Games–Howell.
- Firm data: turnover and emissions are often very skewed → think Kruskal–Wallis or a log.

Errors that cost marks every year

- Writing H_1 as “all groups are different”
- Stopping after a significant F — without post-hoc tests, you cannot say which groups differ
- Running post-hoc tests after a non-significant ANOVA and “finding” a difference
- Reading Levene backwards — $p > .05$ means the variances are equal
- Using Tukey when Levene is significant — use Games–Howell NEW
- “Planning” a Bonferroni comparison after looking at the means NEW
- Comparing groups with a qualitative outcome — that is a chi-square
- Reporting F without its two degrees of freedom, or without η^2
- Recommending action on a difference that is not significant

Before session 4

1. Choose the quantitative outcome your manager cares most about (profit, spending, emissions. . .).
2. List at least three predictors: continuous variables, plus one qualitative variable you can code 0/1.
3. Plot the outcome against each continuous predictor (Exploration $\rightarrow$ Scatterplot). Linear? Outliers?
4. Read: Navarro & Foxcroft, *Learning Statistics with Jamovi*, chapters on comparing several means and linear regression.

Today: *do groups differ — and which ones?* Next week: *what drives my outcome — and by how many euros?*

Questions?

NGUYEN Anh-Tuan

Session 4: what drives my outcome? Simple & multiple regression